

8. There are 7^5 possible outcomes in the sample space. If the five passengers are to get off at different floors, the first passenger can get off at any one of the seven floors, the second passenger can then get off at any one of the other six floors, etc. Thus, the probability is

$$\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3}{7^5} = \frac{360}{2401}.$$

9. There are $6!$ possible arrangements in which the six runners can finish the race. If the three runners from team A finish in the first three positions, there are $3!$ arrangements of these three runners among these three positions and there are also $3!$ arrangements of the three runners from team B among the last three positions. Therefore, there are $3! \times 3!$ arrangements in which the runners from team A finish in the first three positions and the runners from team B finish in the last three positions. Thus, the probability is $(3!3!)/6! = 1/20$.
10. We can imagine that the 100 balls are randomly ordered in a list, and then drawn in that order. Thus, the required probability in part (a), (b), or (c) of this exercise is simply the probability that the first, fiftieth, or last ball in the list is red. Each of these probabilities is the same $\frac{r}{100}$, because of the random order of the list.
11. In terms of factorials, $P_{n,k} = n!/[k!(n-k)!]$. Since we are assuming that n and $n-k$ are large, we can use Stirling's formula to approximate both of them. The approximation to $n!$ is $(2\pi)^{1/2}n^{n+1/2}e^{-n}$, and the approximation to $(n-k)!$ is $(2\pi)^{1/2}(n-k)^{n-k+1/2}e^{-(n-k)}$. The approximation to the ratio is the ratio of the approximations because the ratio of each approximation to its corresponding factorial converges to 1. That is,

$$\frac{n!}{k!(n-k)!} \approx \frac{(2\pi)^{1/2}n^{n+1/2}e^{-n}}{k!(2\pi)^{1/2}(n-k)^{n-k+1/2}e^{-(n-k)}} = \frac{e^{-k}n^k}{k!} \left(1 - \frac{k}{n}\right)^{-n-k-1/2}.$$

Further simplification is available if one assumes that k is small compared to n , that is $k/n \approx 0$. In this case, the last factor is approximately e^k , and the whole approximation simplifies to $n^k/k!$. This makes sense because, if $n/(n-k)$ is essentially 1, then the product of the k largest factors in $n!$ is essentially n^k .

1.8 Combinatorial Methods

Commentary

This section ends with an extended example called “The Tennis Tournament”. This is an application of combinatorics that uses a slightly subtle line of reasoning.

Solutions to Exercises

1. We have to assign 10 houses to one pollster, and the other pollster will get to canvas the other 10 houses. Hence, the number of assignments is the number of combinations of 20 items taken 10 at a time,

$$\binom{20}{10} = 184,756.$$

2. The ratio of $\binom{93}{30}$ to $\binom{93}{31}$ is $31/63 < 1$, so $\binom{93}{31}$ is larger.

3. Since $93 = 63 + 30$, the two numbers are the same.
4. Let the sample space consist of all subsets (not ordered tuples) of the 24 bulbs in the box. There are $\binom{24}{4} = 10626$ such subsets. There is only one subset that has all four defectives, so the probability we want is $1/10626$.
5. The number is $\frac{4251!}{(97!4154!)} = \binom{4251}{97}$, an integer.
6. There are $\binom{n}{2}$ possible pairs of seats that A and B can occupy. Of these pairs, $n - 1$ pairs comprise two adjacent seats. Therefore, the probability is $\frac{n - 1}{\binom{n}{2}} = \frac{2}{n}$.
7. There are $\binom{n}{k}$ possible sets of k seats to be occupied, and they are all equally likely. There are $n - k + 1$ sets of k adjacent seats, so the probability we want is

$$\frac{n - k + 1}{\binom{n}{k}} = \frac{(n - k + 1)!k!}{n!}.$$

8. There are $\binom{n}{k}$ possible sets of k seats to be occupied, and they are all equally likely. Because the circle has no start or end, there are n sets of k adjacent seats, so the probability we want is

$$\frac{n}{\binom{n}{k}} = \frac{(n - k)!k!}{(n - 1)!}.$$

9. This problem is slightly tricky. The total number of ways of choosing the n seats that will be occupied by the n people is $\binom{2n}{n}$. Offhand, it would seem that there are only two ways of choosing these seats so that no two adjacent seats are occupied, namely:

$$X0X0 \dots 0 \quad \text{and} \quad 0X0X \dots 0X$$

Upon further consideration, however, $n - 1$ more ways can be found, namely:

$$X00X0X \dots 0X, \quad X0X00X0X \dots 0X, \text{ etc.}$$

Therefore, the total number of ways of choosing the seats so that no two adjacent seats are occupied is $n + 1$. The probability is $(n + 1)/\binom{2n}{n}$.

10. We shall let the sample space consist of all subsets (unordered) of 10 out of the 24 light bulbs in the box. There are $\binom{24}{10}$ such subsets. The number of subsets that contain the two defective bulbs is the number of subsets of size 8 out of the other 22 bulbs, $\binom{22}{8}$, so the probability we want is

$$\frac{\binom{22}{8}}{\binom{24}{10}} = \frac{10 \times 9}{24 \times 23} = 0.1630.$$

11. This exercise is similar to Exercise 10. Let the sample space consist of all subsets (unordered) of 12 out of the 100 people in the group. There are $\binom{100}{12}$ such subsets. The number of subsets that contain A and B is the number of subsets of size 10 out of the other 98 people, $\binom{98}{10}$, so the probability we want is

$$\frac{\binom{98}{10}}{\binom{100}{12}} = \frac{12 \times 11}{100 \times 99} = 0.01333.$$

12. There are $\binom{35}{10}$ ways of dividing the group into the two teams. As in Exercise 11, the number of ways of choosing the 10 players for the first team so as to include both A and B is $\binom{33}{8}$. The number of ways of choosing the 10 players for this team so as not to include either A or B (A and B will then be together on the other team) is $\binom{33}{10}$. The probability we want is then

$$\frac{\binom{33}{8} + \binom{33}{10}}{\binom{35}{10}} = \frac{10 \times 9 + 25 \times 24}{35 \times 34} = 0.5798.$$

13. This exercise is similar to Exercise 12. Here, we want four designated bulbs to be in the same group. The probability is

$$\frac{\binom{20}{6} + \binom{20}{10}}{\binom{24}{10}} = 0.1140.$$

14.

$$\begin{aligned}
 \binom{n}{k} + \binom{n}{k-1} &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} \\
 &= \frac{n!}{(k-1)!(n-k)!} \left(\frac{1}{k} + \frac{1}{n-k+1} \right) \\
 &= \frac{n!}{(k-1)!(n-k)!} \cdot \frac{n+1}{k(n-k+1)} \\
 &= \frac{(n+1)!}{k!(n-k+1)!} = \binom{n+1}{k}.
 \end{aligned}$$

15. (a) If we express 2^n as $(1+1)^n$ and expand $(1+1)^n$ by the binomial theorem, we obtain the desired result.
- (b) If we express 0 as $(1-1)^n$ and expand $(1-1)^n$ by the binomial theorem, we obtain the desired result.
16. (a) It is easier to calculate first the probability that the committee will not contain either of the two senators from the designated state. This probability is $\binom{98}{8} / \binom{100}{8}$. Thus, the final answer is

$$1 - \frac{\binom{98}{8}}{\binom{100}{8}} \approx 1 - .08546 = 0.1543.$$

- (b) There are $\binom{100}{50}$ combinations that might be chosen. If the group is to contain one senator from each state, then there are two possible choices for each of the fifty states. Hence, the number of possible combinations containing one senator from each state is 2^{50} .
17. Call the four players A, B, C, and D. The number of ways of choosing the positions in the deck that will be occupied by the four aces is $\binom{52}{4}$. Since player A will receive 13 cards, the number of ways of choosing the positions in the deck for the four aces so that all of them will be received by player A is $\binom{13}{4}$. Similarly, since player B will receive 13 other cards, the number of ways of choosing the positions for the four aces so that all of them will be received by player B is $\binom{13}{4}$. A similar result is true for each of the other players. Therefore, the total number of ways of choosing the positions in the deck for the four aces so that all of them will be received by the same player is $4 \binom{13}{4}$. Thus, the final probability is $4 \binom{13}{4} / \binom{52}{4}$.
18. There are $\binom{100}{10}$ ways of choosing ten mathematics students. There are $\binom{20}{2}$ ways of choosing two

students from a given class of 20 students. Therefore, there are $\binom{20}{2}^5$ ways of choosing two students from each of the five classes. So, the final answer is $\binom{20}{2}^5 / \binom{100}{10} \approx 0.0143$.

19. From the description of what counts as a collection of customer choices, we see that each collection consists of a tuple $(m_1, \dots, m_n)$, where m_i is the number of customers who choose item i for $i = 1, \dots, n$. Each m_i must be between 0 and k and $m_1 + \dots + m_n = k$. Each such tuple is equivalent to a sequence of $n + k - 1$ 0's and 1's as follows. The first m_1 terms are 0 followed by a 1. The next m_2 terms are 0 followed by a 1, and so on up to m_{n-1} 0's followed by a 1 and finally m_n 0's. Since $m_1 + \dots + m_n = k$ and since we are putting exactly $n - 1$ 1's into the sequence, each such sequence has exactly $n + k - 1$ terms. Also, it is clear that each such sequence corresponds to exactly one tuple of customer choices. The numbers of 0's between successive 1's give the numbers of customers who choose that item, and the 1's indicate where we switch from one item to the next. So, the number of combinations of choices is the number of such sequences: $\binom{n + k - 1}{k}$.

20. We shall use induction. For $n = 1$, we must prove that

$$x + y = \binom{1}{0} x^0 y^1 + \binom{1}{1} x^1 y^0.$$

Since the right side of this equation is $x + y$, the theorem is true for $n = 1$. Now assume that the theorem is true for each $n = 1, \dots, n_0$ for $n_0 \geq 1$. For $n = n_0 + 1$, the theorem says

$$(x + y)^{n_0+1} = \sum_{k=0}^{n_0+1} \binom{n_0+1}{k} x^k y^{n_0+1-k}. \quad (\text{S.1.2})$$

Since we have assumed that the theorem is true for $n = n_0$, we know that

$$(x + y)^{n_0} = \sum_{k=0}^{n_0} \binom{n_0}{k} x^k y^{n_0-k}. \quad (\text{S.1.3})$$

We shall multiply both sides of (S.1.3) by $x + y$. We then need to prove that $x + y$ times the right side of (S.1.3) equals the right side of (S.1.2).

$$\begin{aligned} (x + y)(x + y)^{n_0} &= (x + y) \sum_{k=0}^{n_0} \binom{n_0}{k} x^k y^{n_0-k} \\ &= \sum_{k=0}^{n_0} \binom{n_0}{k} x^{k+1} y^{n_0-k} + \sum_{k=0}^{n_0} \binom{n_0}{k} x^k y^{n_0+1-k} \\ &= \sum_{k=1}^{n_0+1} \binom{n_0}{k-1} x^k y^{n_0+1-k} + \sum_{k=0}^{n_0} \binom{n_0}{k} x^k y^{n_0+1-k} \\ &= y^{n_0+1} + \sum_{k=1}^{n_0} \left[\binom{n_0}{k-1} + \binom{n_0}{k} \right] x^k y^{n_0+1-k} + x^{n_0+1}. \end{aligned}$$

Now, apply the result in Exercise 14 to conclude that

$$\binom{n_0}{k-1} + \binom{n_0}{k} = \binom{n_0+1}{k}.$$

This makes the final summation above equal to the right side of (S.1.2).

21. We are asked for the number of unordered samples with replacement, as constructed in Exercise 19. Here, $n = 365$, so there are $\binom{365+k}{k}$ different unordered sets of k birthdays chosen with replacement from $1, \dots, 365$.
22. The approximation to $n!$ is $(2\pi)^{1/2}n^{n+1/2}e^{-n}$, and the approximation to $(n/2)!$ is $(2\pi)^{1/2}(n/2)^{n/2+1/2}e^{-n/2}$. Then

$$\frac{n!}{(n/2)!^2} \approx \frac{(2\pi)^{1/2}n^{n+1/2}e^{-n}}{[(2\pi)^{1/2}(n/2)^{n/2+1/2}e^{-n/2}]^2} = (2\pi)^{-1/2}2^{n+1}n^{-1/2}.$$

With $n = 500$, the approximation is $e^{343.24}$, too large to represent on a calculator with only two-digit exponents. The actual number is about 1/20 of 1% larger.

1.9 Multinomial Coefficients

Commentary

Multinomial coefficients are useful as a counting method, and they are needed for the definition of the multinomial distribution in Sec. 5.9. They are not used much elsewhere in the text. Although this section does not have an asterisk, it could be skipped (together with Sec. 5.9) if one were not interested in the multinomial distribution or the types of counting arguments that rely on multinomial coefficients.

Solutions to Exercises

1. We have three types of elements that need to be assigned to 21 houses so that exactly seven of each type are assigned. The number of ways to do this is the multinomial coefficient

$$\binom{21}{7, 7, 7} = 399,072,960.$$

2. We are asked for the number of arrangements of four distinct types of objects with 18 or one type, 12 of the next, 8 of the next and 12 of the last. This is the multinomial coefficient $\binom{50}{18, 12, 8, 12}$.
3. We need to divide the 300 members of the organization into three subsets: the 5 in one committee, the 8 in the second committee, and the 287 in neither committee. There are $\binom{300}{5, 8, 287}$ ways to do this.
4. There are $\binom{10}{3, 3, 2, 1, 1}$ arrangements of the 10 letters of four distinct types. All of them are equally likely, and only one spells statistics. So, the probability is $1/\binom{10}{3, 3, 2, 1, 1} = 1/50400$.
5. There are $\binom{n}{n_1, n_2, n_3, n_4, n_5, n_6}$ many ways to arrange n_j j 's (for $j = 1, \dots, 6$) among the n rolls. The number of possible equally likely rolls is 6^n . So, the probability we want is $\frac{1}{6^n} \binom{n}{n_1, n_2, n_3, n_4, n_5, n_6}$.